EEE-1221

Electronics

<u>FINAL SUGGESTION</u>

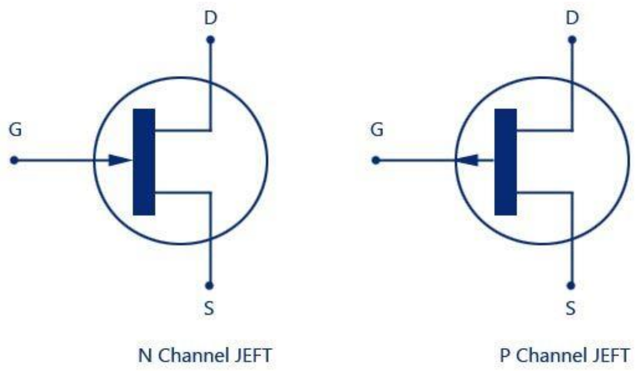
1. Write down the major differences between junction field effect transistor (JFET) and bipolar transistor (BJT)?

Parameter	JFET	BJT
Carrier Type	Majority carriers (electrons in n-channel, holes in p-channel)	Both majority and minority carriers
Control Mechanism	Voltage-controlled	Current-controlled
Input Impedance	Very high	Relatively low
Noise Level	Low	Higher compared to JFET
Thermal Stability	Better	Less stable
Switching Speed	Generally slower	Faster
Power Consumption	Lower	Higher
Gain	Voltage gain	Current gain

2. Draw the electronic symbol of n-channel and p-channel JFET? Describe the construction details and working principle of n-channel JFET.

Symbols:

- n-channel JFET
- p-channel JFET

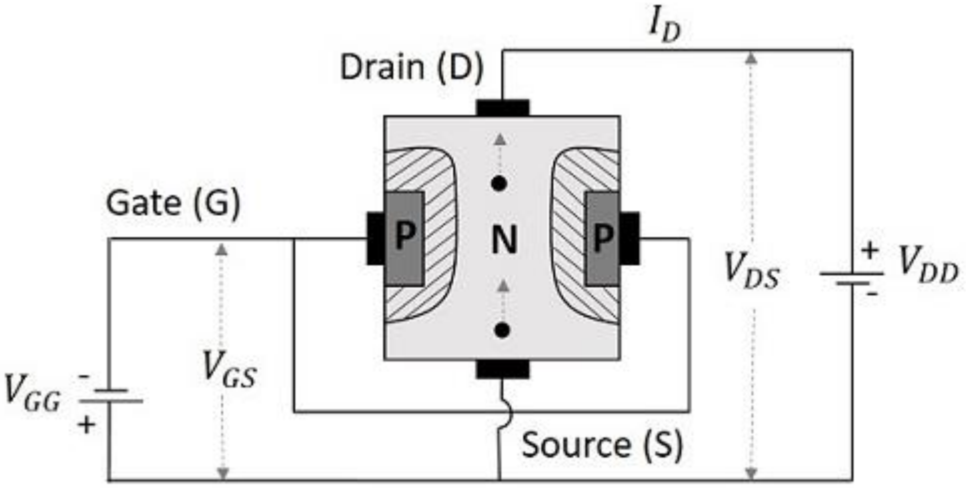


JFET-N-Channel and P-channel Schematic Symbol

Construction and Working Principle of n-channel JFET:

Construction:

The n-channel JFET consists of a long channel of n-type semiconductor material with a gate terminal made from p-type material diffused on both sides of the n-channel, forming a p-n junction.



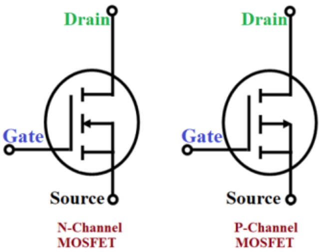
Working Principle:

When no voltage is applied to the gate (V_GS = 0), the channel conducts maximum current (I_DSS). When a negative voltage is applied to the gate, the p-n junctions become reverse-biased, creating a depletion region that narrows the channel. This reduces the current flow. The more negative the gate voltage, the less the current that flows from drain to source.

3. Explain the enhancement-mode and depletion-mode operation of metal oxide semiconductor field effect transistor (MOSFET) with proper circuit diagrams.

Enhancement-Mode MOSFET:

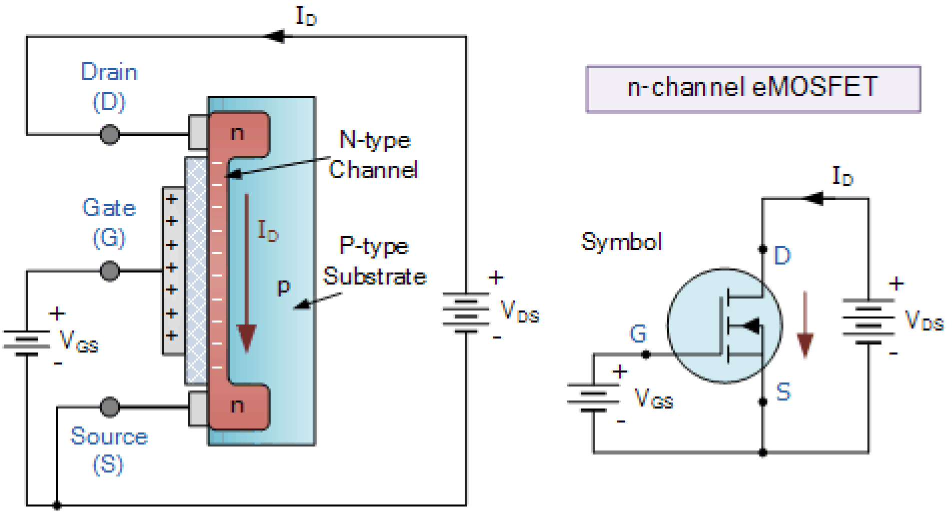
Symbol:



Operation:

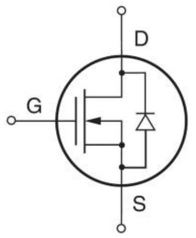
In enhancement-mode MOSFETs, no current flows between the drain and source when the gate-source voltage (V_GS) is zero. A positive gate voltage creates an electric field that enhances the channel conductivity, allowing current to flow.

Circuit Diagram:



Depletion-Mode MOSFET:

Symbol:

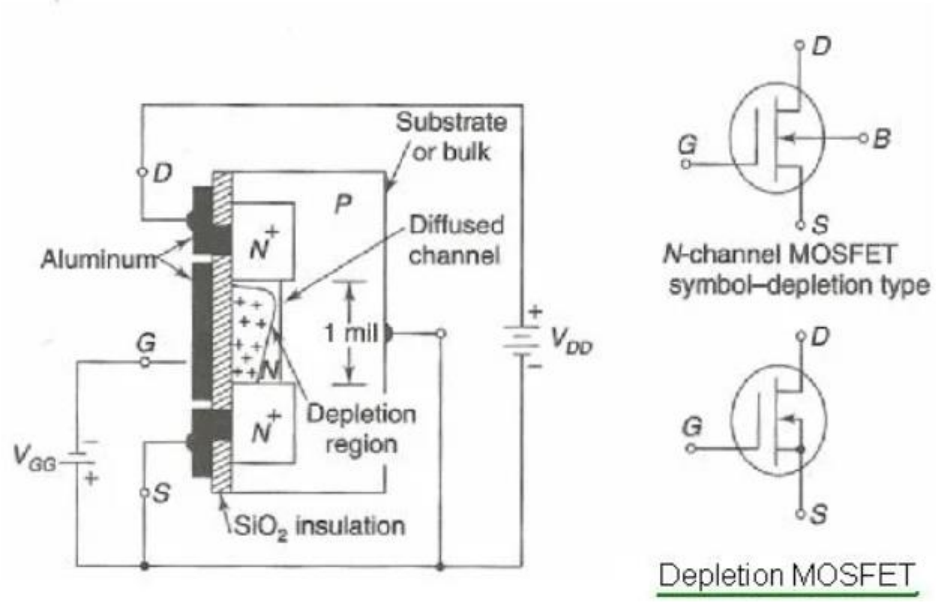


Operation:

In depletion-mode MOSFETs, the device conducts when V_GS = 0. A negative gate voltage depletes the channel of charge carriers, reducing current flow. A positive gate voltage enhances the channel conductivity.

Circuit Diagram:

Depletion MOSFET



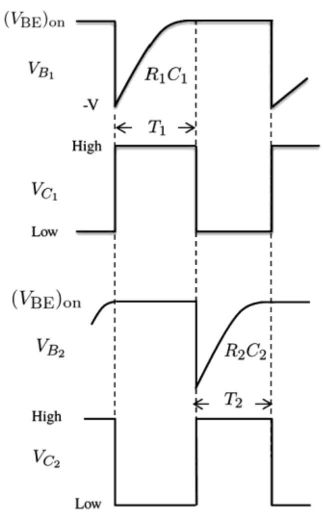
4. What is a multivibrator? Mention different types of multivibrators with proper wave shapes. With neat diagrams, explain the working of a astable multivibrator.

A multivibrator is an electronic circuit used to implement a variety of simple two-state systems such as oscillators, timers, and flip-flops.

Types and Wave Shapes:

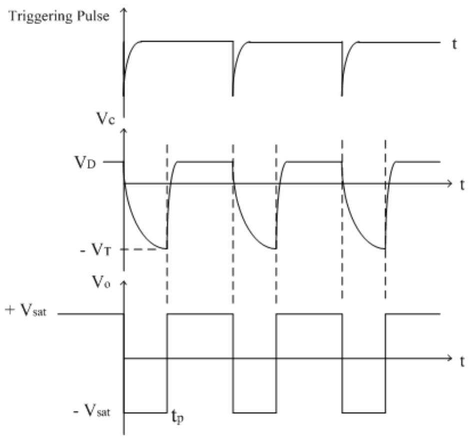
Astable Multivibrator: Produces a continuous square wave output without any stable state.

Waveform:



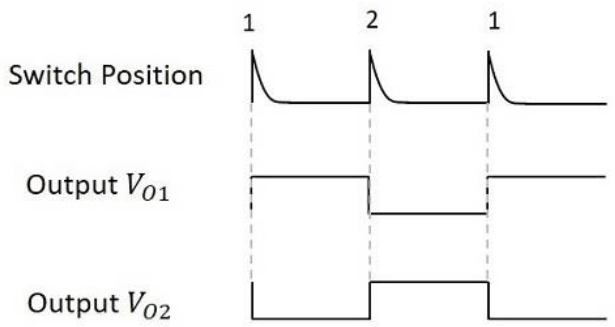
Monostable Multivibrator: Has one stable state and one quasi-stable state. Triggering it causes it to switch to the quasi-stable state for a duration before returning to the stable state.

Waveform:



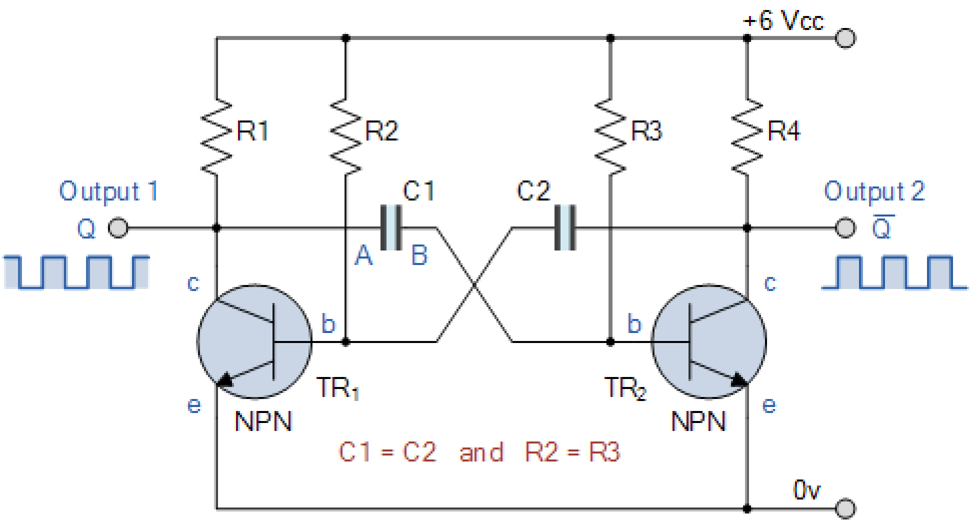
Bistable Multivibrator (Flip-Flop): Has two stable states and requires an external trigger to switch between these states.

Waveform:



Astable Multivibrator Working:

Diagram:



Working:

The circuit consists of two transistors, two capacitors, and four resistors. When the circuit is powered on, one transistor turns on faster than the other, causing one capacitor to charge while the other discharges. This creates a feedback loop where the transistors alternate states, producing a continuous square wave.

5. Explain the essential conditions for the operation of an oscillator. With the help of a neat diagram, describe the circuit operation of a Hartley oscillator.

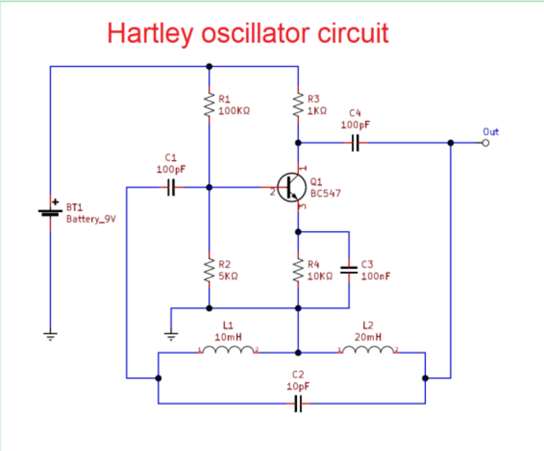
Essential Conditions for Oscillator Operation:

Positive Feedback: The feedback must be in phase with the input signal.

Loop Gain: The product of the amplifier gain and feedback network should be equal to one.

Hartley Oscillator Circuit:

Diagram:



Operation:

The Hartley oscillator uses an LC circuit for frequency determination. The feedback is taken from the tap on the coil. The transistor amplifies the signal, and the feedback through the coil generates sustained oscillations. The frequency of oscillation is determined by the inductance and capacitance values in the circuit.

6. What do you mean by negative feedback? Show the input impedance of an amplifier increases due to negative feedback.

Negative Feedback:

Negative feedback occurs when a portion of the output signal is inverted and fed back to the input, reducing the gain but increasing stability, bandwidth, and input impedance.

Input Impedance Increase:

Negative feedback increases the input impedance of an amplifier by reducing the effective input current for a given input voltage. This can be shown with the following relationship:

$$Z_{in} = \frac{Z_{in(no\ feedback)}}{1 - A\beta}$$

where $Z_{in(no\ feedback)}$ is the input impedance without feedback, A is the gain, and β is the feedback factor.

7. Describe the switching action of the transistor by showing the 'OFF' region, 'ON' region, and 'active' regions on its output characteristics.

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8. Write short notes on: (i) Peak Detector circuit (ii) Comparators (iii) Precision Rectifier.

(i) Peak Detector Circuit:

A peak detector circuit captures the peak value of an input signal. It typically uses a diode, capacitor, and an operational amplifier. The diode allows current to charge the capacitor to the peak voltage of the input signal. Once charged, the capacitor retains this voltage as the peak value.

(ii) Comparators:

A comparator is an operational amplifier configured to compare two voltages. It outputs a high voltage if the positive input is higher than the negative input, and a low voltage otherwise. Comparators are used in applications such as zero-crossing detectors and level shifters.

(iii) Precision Rectifier:

A precision rectifier, or super diode, uses an operational amplifier to rectify small input voltages without the voltage drop associated with diodes. It is used in signal processing applications where accurate rectification of small signals is needed.